Campo elettrico conservativo

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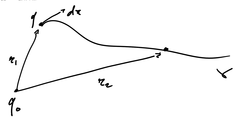
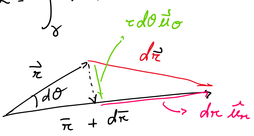


q si sposta con $v \to 0$

$$dL = \vec{F} \cdot dr$$

$$L = \int_\gamma \vec{F}\, d\vec{r}$$



$$d\vec{r} = dr\,\hat{u}_r + r\,d\theta\,\hat{u}_\theta$$

$$\vec{F} = \frac{q_0 q}{4\pi\varepsilon_0 r^2}\,\hat{u}_r$$

$u_r \cdot u_r = 1$

$u_r \cdot u_\theta = 0$

$$L = \int_{r_1}^{r_2} \frac{q_0 q}{4\pi\varepsilon_0 r^2}\, u_r \left(dr\,\hat{u}_r + \cancel{r\,d\theta\,\hat{u}_\theta}\right)$$

$$= \frac{q_0 q}{4\pi\varepsilon_0}\left(-\frac{1}{r}\right)_{r_1}^{r_2}$$

$$= \frac{q_0 q}{4\pi\varepsilon_0}\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

Il lavoro non dipende da $\gamma \Rightarrow \vec{E}$ campo conservativo

Potenziale:

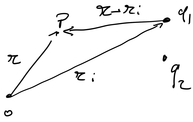
$$V = \frac{U}{q} = \int_r^{rif} \vec{E} \cdot d\vec{r} \qquad V(rif) = 0$$

$$\Delta V = V(B) - V(A) = -\int_A^B \vec{E} \cdot d\vec{r}$$

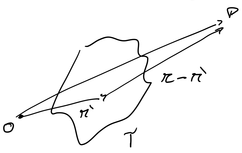
$$dV = -\vec{E} \cdot dr \quad \text{ma} \quad dV = \nabla V \cdot dr$$

$$\Rightarrow \vec{\nabla} V = -\vec{E}$$

Avendo posto $V(\infty) = 0$



- N cariche $\quad V = \dfrac{1}{4\pi\varepsilon_0} \displaystyle\sum_i^N \frac{q}{|r - r_i|}$



- ρ: $\quad V(r) = \dfrac{1}{4\pi\varepsilon_0} \displaystyle\iiint_\tau \frac{\rho\, d\tau'}{|r - r'|}$

- σ: $\quad V(r) = \dfrac{1}{4\pi\varepsilon_0} \displaystyle\iint_\Sigma \frac{\sigma\, dS'}{|r - r'|}$

- λ: $\quad V(r) = \dfrac{1}{4\pi\varepsilon_0} \displaystyle\int_\gamma \frac{\lambda\, dl'}{|r - r'|}$

Energia potenziale di sistemi di cariche

$$U = \frac{1}{2} \sum_i^N q_i V_i$$

Nel continuo

$$U = \frac{1}{2} \int \rho V\, d\tau$$

Energia del campo elettrico

$$U = \frac{1}{2} \iiint_{\tau_\infty} \varepsilon_0 E^2\, d\tau$$

$$u = \frac{1}{2}\varepsilon_0 E^2$$

III eq Maxwell caso statico

$$E \text{ cons} \Rightarrow \oint_{\gamma = \partial\Sigma} \vec{E} \cdot dl = 0$$

Per il tra rotaro

$$0 = \iint_\Sigma (\nabla \times E) \cdot n\, dS$$

$$\Rightarrow \vec{\nabla} \times \vec{E} = 0$$